Area scale factor

1

a 100 **b** $\frac{9}{100}$ **c** $\frac{1}{9}$

2

a 2 **b** $\frac{1}{2}$ **c** 6 **d** 24 units^2

e 4 units^2

3

a $x = 36 \text{ cm}^2$ **b** $x = 504 \text{ cm}^2$

4

a $\frac{3}{7}$ **b** $\frac{9}{49}$

5

a 2 cm **b** 36 cm^2 **c** 9

6 270 cm^2

7 324 cm^2

8 600 cm^2

9 The area will increase by a factor of 9.

10

a $21, 30, 48$ **b** 99 cm

11

a 128 cm^2 **b** 9 cm

12

a The value of A will be doubled. **b** A will increase by a factor of 4.

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13

a A will increase by a factor of 9.

b A will decrease by a factor of 16.

Volume scale factor

14

a 2

b 8

15

a 4

b 64

c $54\,528\ \text{cm}^3$

Surface area scale factor

16

a 5

b 25

c $177\ \text{cm}^2$

17

a $\frac{8}{7}$

b $\frac{512}{343}$

18

a 4

b 2

c $26\ \text{mm}$

19

a $\frac{11}{5}$

b $\frac{121}{25}$

20

a 1.4

b 1.96

c 2.744

Applications

21 $4\ \text{cm}$

22 $15\ \text{cm}$

23

a 6

b 36

c $1.4\ \text{m}^2$

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24
a $60\ cm^2$ **b** $240\ cm^2$ **c** 2 **d** 4

25 3

26
a 2 : 1 **b** 8 : 1 **c** 8

27
a 5 **b** 25 **c** $9375\ cm^2$

28
a 9 **b** 27

29
a 2 : 1 **b** 4 : 1 **c** 8 : 1 **d** 16 : 1

30
a 4 **b** 16 **c** 24 L

31
a 1 : 400 **b** 1 : 8000 **c** 7.2 L **d** 6 mL

32
a 1 : 20 **b** 1 : 400 **c** 1 : 8000

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